
Discrete Fourier Transform & Its Applications

Math – 4, Fall 2024-2025

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1 Introduction

The knowledge of the acquisition and inspection of one-dimensional and two-dimensional signals, namely audio and images, has been made feasible due to the emergence of a number of orthogonal transforms. The Fourier transform is a highly significant tool that has been frequently used for the analysis and processing of one-dimensional and two-dimensional signals, as well as for gaining insights into their characteristics and origins.

The process of decomposing a one-dimensional or two-dimensional signal, such as audio or image data, involves utilizing the Fourier transform to identify the constituent sinusoids with varying periods that represent the temporal or spatial frequencies within the signal. The magnitudes of these frequencies collectively form the frequency spectrum of the signal. The inverse Fourier transform incorporates the synthesis of the signal by aggregating its constituent frequencies.

In this chapter, fortunately, we will learn:

- What is signal
- What is digital signal
- What is image
- What is digital image
- 1-D DFT & its implementation in DSP (basic)
- 2-D DFT & its implementation in DIP (basic)

2 Digital Signal & Image

A signal is any physical process that varies over time. Analog signals are waveforms that exhibit continuity in the temporal domain and possess the ability to assume a continuous spectrum of amplitude values. One example of an analog signal is a voltage that may be administered to an oscilloscope, leading to a continuous representation in relation to time. On the other hand, a digital signal is characterized by the quantization of its independent time variable, resulting in knowing the signal's value only at discrete time intervals. Therefore, a digital signal is characterized by a sequence of values rather than a continuous waveform. The concept of the analog and digital signal is depicted in Fig. 1.

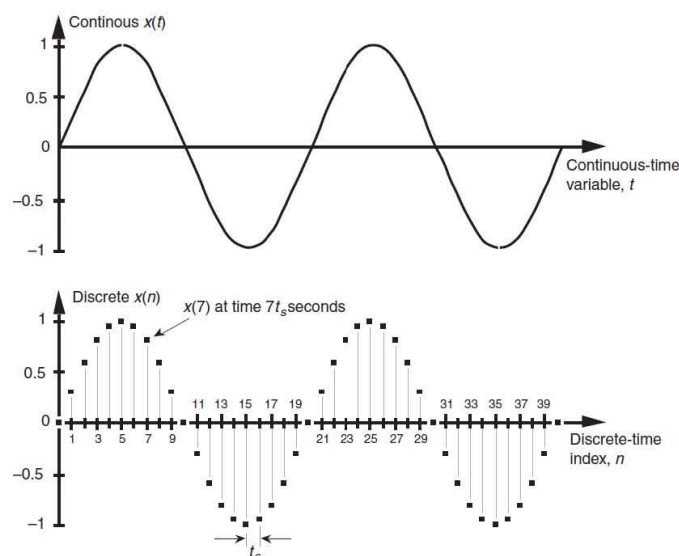


Figure 1: An analog signal and its digital version.

Next, an analog image is a two-dimensional waveform. Unlike an analog signal, it is a function of two variables in space (length). For example, the two-dimensional projection of the

three-dimensional scene around us. Mathematically, a digital image is a function $I(m, n)$ of independent spatial variables m and n quantized so that we know only the value of $I(m, n)$ at discrete length in space.

2.1 1-D Digital Signal (Audio)

Now, we will learn how to discretize an analog signal. Let us consider the analog signal $f(t)$ on the time interval $0 \leq t \leq b$. Say, the digital representation of $f(t)$ is $F(n)$, where n is the integer index belongs to $\{0\} \cup \mathbb{N}$. First, we have to fix the sample size in the digital signal $F(n)$, which is N . Thus, the n varies from 0 to $N - 1$. Then, the time interval t_s between two successive samples can be determined as $t_s = \frac{b}{N-1}$. The m th sample of the digital signal $F(n)$, therefore, can be computed following $F(n) = f(n \times t_s)$, where $n = m - 1$. Now, the digital signal $F(n)$ can be computed by computing its samples as:

The first sample ($n = 0$) is $F(n) = f(n \times t_s) \Rightarrow F(0) = f(0 \times t_s)$.

The second sample ($n = 1$) is $F(n) = f(n \times t_s) \Rightarrow F(1) = f(1 \times t_s)$.

The third sample ($n = 2$) is $F(n) = f(n \times t_s) \Rightarrow F(2) = f(2 \times t_s)$.

$\vdots$

The N th sample ($n = N - 1$) is $F(n) = f(n \times t_s) \Rightarrow F(N - 1) = f((N - 1) \times t_s)$.

Hence, the digital signal is $F(n) = \{F(0), F(1), F(2), \dots, F(N - 1)\}$.

Note: *In a digital signal, the values of the samples are also quantized. It is skipped here.*

Example 1. Find the digital signal $F(n)$ with 6 samples of the unit ramp signal

$$f(t) = \begin{cases} 0 & \text{for } t < 0 \\ t & \text{for } 0 \leq t \leq 2 \end{cases}.$$

Solution: Here, $N = 6$ and $b = 2$.

Thus, $t_s = \frac{b}{N-1} = \frac{2}{5} = 0.4$, and $n = 0, 1, 2, \dots, 5$.

For $n = 0$, $F(0) = f(0 \times 0.4) = 0$.

For $n = 1$, $F(1) = f(1 \times 0.4) = 0.4$.

For $n = 2$, $F(2) = f(2 \times 0.4) = 0.8$.

For $n = 3$, $F(3) = f(3 \times 0.4) = 1.2$.

For $n = 4$, $F(4) = f(4 \times 0.4) = 1.6$.

For $n = 5$, $F(5) = f(5 \times 0.4) = 2$.

The digital unit ramp function is $F(n) = \{F(0), F(1), F(2), F(3), F(4), F(5)\} = \{0, 0.4, 0.8, 1.2, 1.6, 2\}$.

Example 2. Find the sixth sample of the digital signal $F(n)$ of the unit step signal

$$f(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } 0 \leq t \leq 3 \end{cases} \quad \text{with the time interval } t_s = 0.5 \text{ if it exists. Also, find the eighth sample if it exists.}$$

Solution: Here, the time period is $t_s = 0.5$ and $b = 3$.

Thus, the number of samples $N = \frac{b}{t_s} + 1 = \frac{3}{0.5} + 1 = 6 + 1 = 7$.

So, the sixth sample exists.

For the sixth sample $n = 5$, and $F(5) = f(5 \times t_s) = f(5 \times 0.5) = f(2.5) = 1$.

Since the total number of samples is 7, the eighth sample of $F(n)$ does not exist.

2.2 2-D Digital Signal (Image)

At this time, we will learn how to form a digital image from an analog one. Say, we have the analog image $f(x, y)$ on the spatial intervals $0 \leq x \leq a$ and $0 \leq y \leq b$. Let the digital representation of $f(x, y)$ is $I(m, n)$, where m and n are the integer indices belong to $\{0\} \cup \mathbb{N}$. First, we have to fix the sample size in the digital image $I(m, n)$, which is $M \times N$. Thus, m and n vary from 0 to $M - 1$ and $N - 1$, respectively. Then the spatial interval x_s (and y_s) between two successive samples can be determined by $x_s = \frac{a}{M-1}$ ($y_s = \frac{b}{N-1}$). It is noted here that $x - s = y_s$, and a sample of a digital image is known as a pixel. Therefore, the pixel at the position (m, n) of the digital image $I(m, n)$ can be computed as $I(m, n) = f(m \times x_s, n \times y_s)$. Consider a digital image $I(m, n)$ of size $(M - 1) \times (N - 1)$. So, $I(m, n)$ can be computed by computing its all the nine pixels as:

At the position $(m, n) = (0, 0)$, $I(m, n) = f(m \times x_s, n \times y_s) \Rightarrow I(0, 0) = f(0 \times x_s, 0 \times y_s)$.

At the position $(m, n) = (0, 1)$, $I(m, n) = f(m \times x_s, n \times y_s) \Rightarrow I(0, 1) = f(0 \times x_s, 1 \times y_s)$.

... ..

At the position $(m, n) = (0, N - 1)$, $I(m, n) = f(m \times x_s, n \times y_s) \Rightarrow I(0, N - 1) = f(0 \times x_s, N - 1 \times y_s)$.

At the position $(m, n) = (1, 0)$, $I(m, n) = f(m \times x_s, n \times y_s) \Rightarrow I(1, 0) = f(1 \times x_s, 0 \times y_s)$.

At the position $(m, n) = (1, 1)$, $I(m, n) = f(m \times x_s, n \times y_s) \Rightarrow I(1, 1) = f(1 \times x_s, 1 \times y_s)$.

... ..

At the position $(m, n) = (1, N - 1)$, $I(m, n) = f(m \times x_s, n \times y_s) \Rightarrow I(1, N - 1) = f(1 \times x_s, N - 1 \times y_s)$.

⋮

⋮

At the position $(m, n) = (M - 1, 0)$, $I(m, n) = f(m \times x_s, n \times y_s) \Rightarrow I(M - 1, 0) = f(M - 1 \times x_s, 0 \times y_s)$.

At the position $(m, n) = (M - 1, 1)$, $I(m, n) = f(m \times x_s, n \times y_s) \Rightarrow I(M - 1, 1) = f(M - 1 \times x_s, 1 \times y_s)$.

... ..

At the position $(m, n) = (M - 1, N - 1)$, $I(m, n) = f(m \times x_s, n \times y_s) \Rightarrow I(M - 1, N - 1) = f(M - 1 \times x_s, N - 1 \times y_s)$.

Hence, the digital image is $I(m, n) = \left\{ \begin{array}{cccc} \{I(0, 0), & I(0, 1), & \dots, & I(0, N - 1)\} \\ \{I(1, 0), & I(1, 1), & \dots, & I(1, N - 1)\} \\ & & \ddots & \\ \{I(M - 1, 0), I(M - 1, 1), \dots, I(M - 1, N - 1)\} \end{array} \right\}$.

Note: In a digital image, the values of the pixels are also quantized. It is skipped here.

Example 3. Find the digital image $I(m, n)$ with 3×2 pixel size of the analog image $f(x, y) = x^2 + y^2$, where $0 \leq x \leq 1$ and $0 \leq y \leq 0.5$.

Solution: Here, $M = 3, N = 2, a = 1$ and $b = 0.5$.

Thus, $x_s = \frac{a}{M-1} = \frac{1}{3-1} = \frac{1}{2} = 0.5$ and $y_s = \frac{b}{N-1} = \frac{0.5}{2-1} = \frac{0.5}{1} = 0.5$.

$$I(0, 0) = f(0 \times x_s, 0 \times y_s) = f(0 \times 0.5, 0 \times 0.5) = f(0, 0) = 0^2 + 0^2 = 0.$$

$$I(0, 1) = f(0 \times x_s, 1 \times y_s) = f(0 \times 0.5, 1 \times 0.5) = f(0, 0.5) = 0^2 + 0.5^2 = 0.25.$$

$$I(1, 0) = f(1 \times x_s, 0 \times y_s) = f(1 \times 0.5, 0 \times 0.5) = f(0.5, 0) = 0.5^2 + 0^2 = 0.25.$$

$$I(1, 1) = f(1 \times x_s, 1 \times y_s) = f(1 \times 0.5, 1 \times 0.5) = f(0.5, 0.5) = 0.5^2 + 0.5^2 = 0.5.$$

$$I(2, 0) = f(2 \times x_s, 0 \times y_s) = f(2 \times 0.5, 0 \times 0.5) = f(1, 0) = 1^2 + 0^2 = 1.$$

$$I(2, 1) = f(2 \times x_s, 1 \times y_s) = f(2 \times 0.5, 1 \times 0.5) = f(1, 0.5) = 0.5^2 + 0.5^2 = 1.25.$$

Hence, the digital image is $I(m, n) = \left\{ \begin{array}{cc} I(0, 0) & I(0, 1) \\ I(1, 0) & I(1, 1) \\ I(2, 0) & I(2, 1) \end{array} \right\} = \begin{array}{|c|c|} \hline 0 & 0.25 \\ \hline 0.25 & 0.5 \\ \hline 1 & 1.25 \\ \hline \end{array}$

Example 4. Find the pixel value $I(2, 2)$ of the digital image $I(m, n)$ with spatial intervals $x_s = 0.1$ and $y_s = 0.1$ of the analog image $f(x, y) = \frac{x^3}{y^2}$, where $0 \leq x \leq 2$ and $0 \leq y \leq 2$. Also, find the pixel size of $I(m, n)$.

Solution: Here, $a = 2, b = 2$, and $x_s = y_s = 0.1$.

Thus, $M = \frac{a}{x_s} + 1 = \frac{2}{0.1} + 1 = 20 + 1 = 21$ and $N = \frac{b}{y_s} + 1 = \frac{2}{0.1} + 1 = 20 + 1 = 21$.

So, the pixel size of $I(m, n)$ is $M \times N = 21 \times 21$.

$$\text{Also, } I(2, 2) = f(2 \times x_s, 2 \times y_s) = f(2 \times 0.1, 2 \times 0.1) = f(0.2, 0.2) = \frac{0.2^3}{0.2^2} = 0.2.$$

Note: The real-time image formulation is not an easy task. For simplicity, in this text, an analog image is described in terms of a mathematical (algebraic) function.

3 Discrete Fourier Transform (DFT)

The digital signals converted from analog signals are either in temporal or spatial domain. The frequency domain representation of the signals is required for many important signal

processing tasks, such as signal reconstruction, noise reduction. The DFT is one of the sophisticated procedures to transform the spatial and temporal domain signals into the frequency domain.

3.1 1-D DFT

1-D DFT is used to transform a digital signal in the temporal domain to the frequency domain. Let $F(n) = \{F(0), F(1), F(2), \dots, F(N-1)\}$ presents a digital signal with sample size N , which needs to be transformed into the frequency domain. The 1-D DFT of $F(n)$ is $\mathbb{F}(u)$ which can be given by

$$\mathbb{F}(u) = \frac{1}{N} \sum_{n=0}^{N-1} F(n) \times e^{-i \frac{2\pi nu}{N}},$$

where $i = \sqrt{-1}$ and u is the frequency domain index running from 0 to $N-1$. Each $\mathbb{F}(u)$ for $u = 0, 1, 2, \dots, N-1$ represents a frequency component of the digital signal $F(n)$.

Example 5. Transform the digital signal $F(n) = \{0, 0.4, 0.8, 1.2, 1.6, 2\}$ in the temporal domain into the frequency domain.

Solution: Here, the sample size of $F(n)$ is $N = 6$.

Thus, the frequency index u varies from 0 to 5.

And, $F(0) = 0, F(1) = 0.4, F(2) = 0.8, F(3) = 1.2, F(4) = 1.6$, and $F(5) = 2$.

$$\begin{aligned} \mathbb{F}(0) &= \frac{1}{N} \sum_{n=0}^{N-1} F(n) \times e^{-i \frac{2\pi n0}{N}} = \frac{1}{6} \sum_{n=0}^{6-1} F(n) \times e^0 = \frac{1}{6} \sum_{n=0}^5 F(n) \times 1 \\ &= \frac{F(0)+F(1)+F(2)+F(3)+F(4)+F(5)}{6} = \frac{0+0.4+0.8+1.2+1.6+2}{6} = \frac{6}{6} = 1. \end{aligned}$$

$$\begin{aligned} \mathbb{F}(1) &= \frac{1}{N} \sum_{n=0}^{N-1} F(n) \times e^{-i \frac{2\pi n1}{N}} = \frac{1}{6} \sum_{n=0}^{6-1} F(n) \times e^{-i \frac{2\pi n}{6}} = \frac{1}{6} \sum_{n=0}^5 F(n) \times e^{-i \frac{2\pi n}{6}} \\ &= \frac{1}{6} \times \{(F(0) \times e^{-i \frac{2\pi 0}{6}}) + (F(1) \times e^{-i \frac{2\pi 1}{6}}) + (F(2) \times e^{-i \frac{2\pi 2}{6}}) + (F(3) \times e^{-i \frac{2\pi 3}{6}}) + \\ &\quad (F(4) \times e^{-i \frac{2\pi 4}{6}}) + (F(5) \times e^{-i \frac{2\pi 5}{6}})\} = \frac{1}{6}. \\ &= \frac{1}{6} \times \{(0 \times (0 + 0i)) + (0.4 \times (0.2 - 0.34641i)) + (0.8 \times (-0.4 - 0.69282i)) + \\ &\quad (1.2 \times (-1.2 - 1.4696e - 16i)) + (1.6 \times (-0.8 + 1.3856i)) + (2 \times (1 + 1.7321i))\} \\ &= \frac{-1.2000+2.0785i}{6} = -0.2000 + 0.3464i. \end{aligned}$$

Similarly, we get (*The students should work out the following concise answers.*)

$$\mathbb{F}(2) = -0.2000 + 0.1155i.$$

$$\mathbb{F}(3) = -0.2000 - 0.0000i = -0.2.$$

$$\mathbb{F}(4) = -0.2000 - 0.1155i.$$

$$\mathbb{F}(5) = -0.2000 - 0.3464i.$$

Hence, the frequency domain representation of $F(n)$ is $\mathbb{F}(u) =$

$$\{1, -0.2000 + 0.3464i, -0.2000 + 0.1155i, -0.2, -0.2000 - 0.1155i, -0.2000 - 0.3464i\}.$$

Example 6. For a given digital signal $F(n) = \{0, 0.95, 0.59, -0.59, -0.95\}$, find the fifth frequency domain component if it exists. Does the sixth frequency domain component exist?

Solution: Here, the sample size of $F(n)$ is $N = 5$.

Thus, the frequency index u varies from 0 to 4.

And, $F(0) = 0$, $F(1) = 0.95$, $F(2) = 0.59$, $F(3) = -0.59$, and $F(4) = -0.95$.

So, the fifth frequency domain component of $F(n)$ is

$$\begin{aligned} \mathbb{F}(4) &= \frac{1}{N} \sum_{n=0}^{N-1} F(n) \times e^{-i\frac{2\pi n4}{N}} = \frac{1}{5} \sum_{n=0}^4 F(n) \times e^{-i\frac{8\pi n}{5}} \\ &= \frac{1}{5} \times \{(F(0) \times e^{-i\frac{8\pi \cdot 0}{5}}) + (F(1) \times e^{-i\frac{8\pi \cdot 1}{5}}) + (F(2) \times e^{-i\frac{8\pi \cdot 2}{5}}) + (F(3) \times e^{-i\frac{8\pi \cdot 3}{5}}) + \\ &\quad (F(4) \times e^{-i\frac{8\pi \cdot 4}{5}})\} \\ &= \frac{1}{5} \times \{(0 \times (1 + 0i)) + (0.95 \times (0.30902 + 0.95106i)) + \\ &\quad (0.59 \times (-0.80902 + 0.58779i)) + (-0.59 \times (-0.80902 - 0.58779i)) + \\ &\quad (-0.95 \times (0.30902 - 0.95106i))\} \\ &= \frac{1}{5} \times \{(0 + 0i) + (0.29357 + 0.9035i) + (-0.47732 + 0.34679i) + \\ &\quad (0.47732 + 0.34679i) + (-0.29357 + 0.9035i)\} \\ &= \frac{0+2.5006i}{5} = 0 + 0.50012i = 0.5i. \end{aligned}$$

Since $N = 5$, the sixth frequency domain component of $F(n)$ does not exist.

3.2 2-D DFT

2-D DFT is used to transform a digital image in the spatial domain to the frequency domain.

$$\text{Let } I(m, n) = \left\{ \begin{array}{cccc} I(0, 0) & I(0, 1) & \dots & I(0, N-1) \\ I(1, 0) & I(1, 1) & \dots & I(1, N-1) \\ & & \ddots & \\ I(M-1, 0) & I(M-1, 1) & \dots & I(M-1, N-1) \end{array} \right\} \text{ presents a digital im-}$$

age with pixel size $M \times N$, which needs to be transformed into the frequency domain. The 2-D DFT of $I(m, n)$ is $\mathbb{I}(u, v)$ which can be given by

$$\mathbb{I}(u, v) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} F(m, n) \times e^{-2\pi i \left(\frac{mu}{M} + \frac{nv}{N} \right)},$$

where $i = \sqrt{-1}$, and u and v are the frequency domain indices running from 0 to $M-1$ and $N-1$, respectively.

Example 7. Transform the digital image $I(m, n) = \begin{array}{|c|c|} \hline 0 & 127 \\ \hline 128 & 255 \\ \hline \end{array}$ in the spatial domain into the frequency domain.

Solution: Here, the pixel size of $I(m, n)$ is $M \times N = 2 \times 2$.

Thus, the frequency indices u and v varies from 0 to 1 and 0 to 1, respectively.

And, $I(0, 0) = 0$, $I(0, 1) = 127$, $I(1, 0) = 128$, and $I(1, 1) = 255$.

$$\begin{aligned}
\mathbb{I}(0, 0) &= \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} I(m, n) \times e^{-2\pi i \left(\frac{m0}{M} + \frac{n0}{N} \right)} = \frac{1}{2 \times 2} \sum_{m=0}^{2-1} \sum_{n=0}^{2-1} I(m, n) \times e^0 \\
&= \frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 I(m, n) \times 1 = \frac{1}{4} \sum_{m=0}^1 (I(m, 0) + I(m, 1)) \\
&= \frac{I(0,0)+I(0,1)+I(1,0)+I(1,1)}{4} = \frac{0+127+128+255}{4} = \frac{510}{4} = 127.5
\end{aligned}$$

$$\begin{aligned}
\mathbb{I}(0, 1) &= \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} I(m, n) \times e^{-2\pi i \left(\frac{m0}{M} + \frac{n1}{N} \right)} \\
&= \frac{1}{2 \times 2} \sum_{m=0}^{2-1} \sum_{n=0}^{2-1} I(m, n) \times e^{-2\pi i \left(0 + \frac{n}{2} \right)} = \frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 I(m, n) \times e^{-2\pi i \frac{n}{2}} \\
&= \frac{1}{4} \sum_{m=0}^1 \left(I(m, 0) \times e^{-2\pi i \frac{0}{2}} + I(m, 1) \times e^{-2\pi i \frac{1}{2}} \right) \\
&= \frac{1}{4} \sum_{m=0}^1 \left(I(m, 0) \times e^0 + I(m, 1) \times e^{-\pi i} \right) \\
&= \frac{1}{4} \sum_{m=0}^1 \left(I(m, 0) \times 1 + I(m, 1) \times e^{-\pi i} \right) \\
&= \frac{1}{4} \sum_{m=0}^1 \left(I(m, 0) + I(m, 1) \times e^{-\pi i} \right) \\
&= \frac{I(0,0)+I(0,1) \times e^{-\pi i} + I(1,0) + I(1,1) \times e^{-\pi i}}{4} = \frac{0+127 \times e^{-\pi i} + 128 + 255 \times e^{-\pi i}}{4} \\
&= \frac{0+127 \times (-1) + 128 + 255 \times (-1)}{4} = \frac{0-127+128-255}{4} = \frac{-254}{4} = -63.5
\end{aligned}$$

Similarly, we get (*The students should work out the following concise answers.*)

$$\mathbb{I}(1, 0) = -64 - 0i = -64$$

$$\mathbb{I}(1, 1) = 0 + 0i = 0$$

Hence, the frequency domain representation of $I(m, n)$ is $\mathbb{I}(u, v) =$

127.5	-63.5
-64	0

Example 8. For a given digital image $I(m, n) =$

0	100
200	255

, find the frequency domain component $\mathbb{I}(0, 1)$ of $I(m, n)$.

Solution: Here, the pixel size of $I(m, n)$ is $M \times N = 2 \times 2$.

Thus, the frequency indices u and v varies from 0 to 1 and 0 to 1, respectively.

And, $I(0, 0) = 0, I(0, 1) = 100, I(1, 0) = 200$, and $I(1, 1) = 255$.

So, the required frequency domain component is

$$\begin{aligned}
\mathbb{I}(0, 1) &= \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} I(m, n) \times e^{-2\pi i \left(\frac{m0}{M} + \frac{n1}{N} \right)} \\
&= \frac{1}{2 \times 2} \sum_{m=0}^{2-1} \sum_{n=0}^{2-1} I(m, n) \times e^{-2\pi i \left(0 + \frac{n}{2} \right)} = \frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 I(m, n) \times e^{-2\pi i \frac{n}{2}} \\
&= \frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 I(m, n) \times e^{-\pi i n} = \frac{1}{4} \sum_{m=0}^1 \left(I(m, 0) \times e^{-\pi i 0} + I(m, 1) \times e^{-\pi i 1} \right) \\
&= \frac{1}{4} \sum_{m=0}^1 \left(I(m, 0) \times e^0 + I(m, 1) \times e^{-\pi i} \right) \\
&= \frac{1}{4} \sum_{m=0}^1 \left(I(m, 0) \times 1 + I(m, 1) \times e^{-\pi i} \right) = \frac{1}{4} \sum_{m=0}^1 \left(I(m, 0) + I(m, 1) \times e^{-\pi i} \right) \\
&= \frac{I(0,0)+I(0,1) \times e^{-\pi i} + I(1,0) + I(1,1) \times e^{-\pi i}}{4} = \frac{0+100 \times e^{-\pi i} + 200 + 255 \times e^{-\pi i}}{4} \\
&= \frac{0+100 \times (-1) + 200 + 255 \times (-1)}{4} = \frac{0-100+200-255}{4} = \frac{-155}{4} = -38.75.
\end{aligned}$$

4 Exercise 3.7

4.1 1-D Digital Signal (Audio)

Exercise 1. Find the digital signal $F(n)$ with 5 samples of the unit step signal $f(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } 0 \leq t \leq 1 \end{cases}$.

Exercise 2. Find the digital signal $F(n)$ with 11 samples of the wave signal $f(t) = \sin(2\pi f_0 t)$, where $f_0 = 0.1$ and $0 \leq t \leq 10$.

Exercise 3. Find the third sample of the digital signal $F(n)$ of the unit ramp signal $f(t) = \begin{cases} 0 & \text{for } t < 0 \\ t & \text{for } 0 \leq t \leq 2 \end{cases}$ with the time interval $t_s = 0.4$ if it exists. Also, find the ninth sample if it exists.

Exercise 4. Find the twelfth sample of the digital signal $F(n)$ of the wave signal $f(t) = \cos(2\pi f_0 t)$, where $f_0 = 0.2$ and $0 \leq t \leq 5$, with the time interval $t_s = 0.25$ if it exists. Also, find the ninth sample if it exists.

4.2 2-D Digital Signal (Image)

Exercise 5. Find the digital image $I(m, n)$ with 3×3 pixel size of the analog image $f(x, y) = \frac{x+y}{y}$, where $0 \leq x \leq 0.4$ and $0 \leq y \leq 0.4$.

Exercise 6. Find the pixel value $I(3, 4)$ of the digital image $I(m, n)$ with spatial intervals $x_s = 1$ and $y_s = 1$ of the analog image $f(x, y) = \sqrt{(x^2 + y^2)}$, where $0 \leq x \leq 5$ and $0 \leq y \leq 6$. Also, find the pixel size of $I(m, n)$.

4.3 1-D DFT

Exercise 7. Transform the digital signal $F(n) = \{1, 0.31, -0.81, -0.81, 0.31\}$ in the temporal domain into the frequency domain.

Exercise 8. For a given digital signal $F(n) = \{1, 1, 1, 1, 1\}$, find the third frequency domain component if it exists. Does the ninth frequency domain component exist?

4.4 2-D DFT

Exercise 9. Transform the digital image $I(m, n) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ in the spatial domain into the frequency domain.

Exercise 10. For a given digital image $I(m, n) = \begin{bmatrix} 255 & 255 \\ 255 & 255 \end{bmatrix}$, find the frequency domain component $\mathbb{I}(1, 1)$ of $I(m, n)$.

References

1. Tinku Acharya and Ajoy K Ray, “Image Processing: Principles and Applications”, 1st Edition, **Wiley-Interscience** (2005)